Computational Neuroscience — Module 06: Learning — Study Questions

1. What is Hebbian learning? State the rule in your own words.
2. What is the difference between LTP and LTD?
3. How do NMDA receptors act as coincidence detectors?
4. Why is the NMDA receptor's requirement for both presynaptic and postsynaptic activity important for learning?
5. Describe the three conditions for dopamine neuron firing: unexpected reward, expected reward, missing reward.
6. How is the dopamine reward prediction error signal similar to prediction error in Active Inference?
7. What brain region do dopamine neurons primarily project to for reward-based learning?
8. What is the role of the hippocampus in memory formation?
9. What are sharp-wave ripples and what role do they play in memory consolidation?
10. How does hippocampal replay during sleep implement model updating?
11. What are place cells? What is a cognitive map?
12. What is STDP? How does the timing of pre- and post-synaptic spikes determine whether LTP or LTD occurs?
13. Why is the ~20-40ms timing window of STDP significant?
14. How is STDP related to prediction at the synaptic level?
15. What happens at the molecular level when LTP is induced?
16. How does the cerebellum's learning mechanism (climbing fibers) compare to hippocampal learning?
17. Can learning be maladaptive? Give a neural example (e.g., addiction, PTSD).
18. How does dopamine-based learning relate to addiction? What goes wrong?
19. What is the difference between episodic memory (hippocampus) and procedural memory (cerebellum/basal ganglia)?
20. How does the computational neuroscience view of learning complement the cognitive science view (Unit 1, Module 06)?